

Quantium Chip Sales Analysis Report

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Prepared for

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1 Executive Summary

This report explores chip sales data across customer segments and time, using merged transaction and customer behavior data. The analysis reveals which brands, timeframes, and demographics drive revenue for Quantum's chip category.

2 Key Insights

2.1 Top 5 Brands by Total Sales

Kettle leads with \$403,588, followed by Smiths (\$273,778), Doritos (\$267,806), Pringles (\$221,438), and Old El Paso (\$91,615).

2.2 Monthly Sales Trends

Strongest month: December 2018, with total sales of \$170,786. Lowest month: June 2018 (partial month), with only \$27,358.

Overall sales trend is relatively stable with peaks during holiday months (November–December) and small dips mid-year.

2.3 Customer Premium Segments

Mainstream customers generate the highest revenue: \$750,744 (approximately 39% of total). Budget segment contributes \$676,212, while Premium accounts for \$507,459.

Total across segments: \$1.93 million.

2.4 Customer Life Stages

Top 3 life stages by total sales: Older Singles/Couples: \$402,427 Retirees: \$366,471 Older Families: \$353,767
Lowest spenders are New Families at \$50,433.

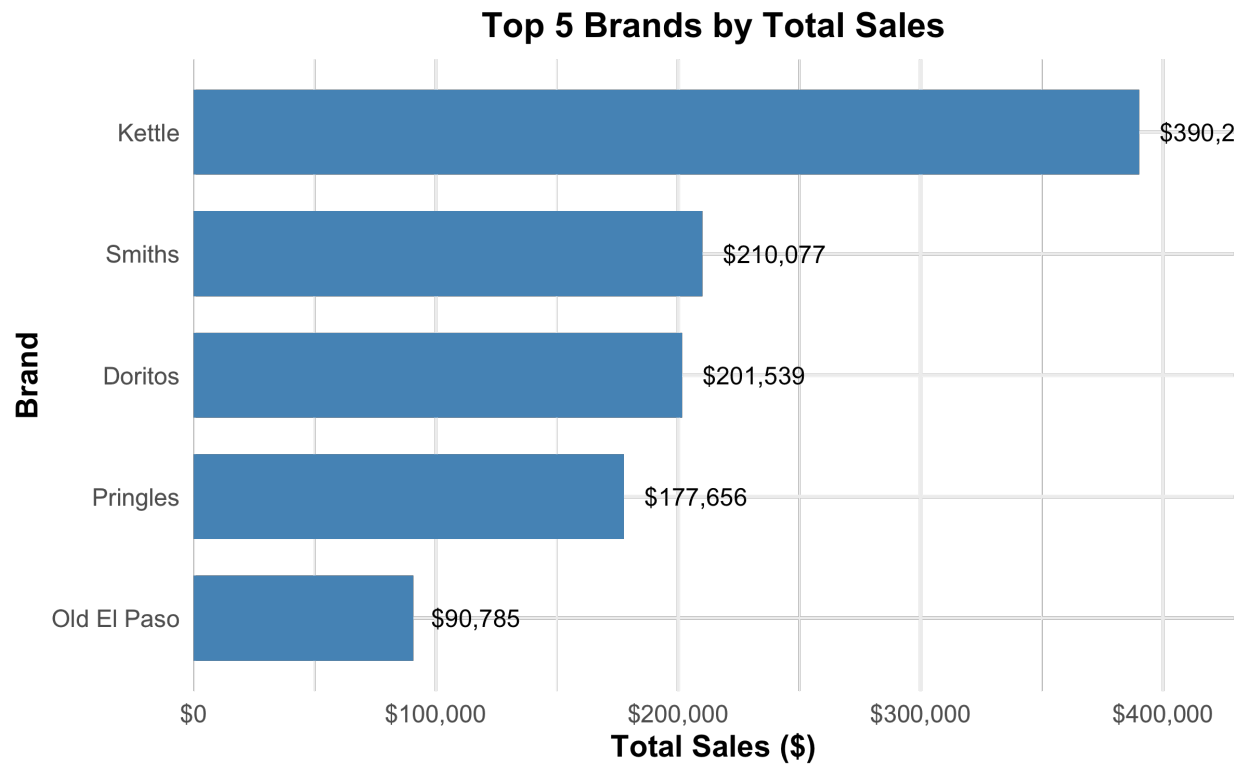
2.5 Weekly Sales Pattern

Weekly sales average: approximately \$37,200. Highest weekly sales: \$44,332 (December 23–29, 2018). Lowest weekly sales: \$30,514 (June 4–10, 2018), likely due to partial data.

3 Data Visualizations

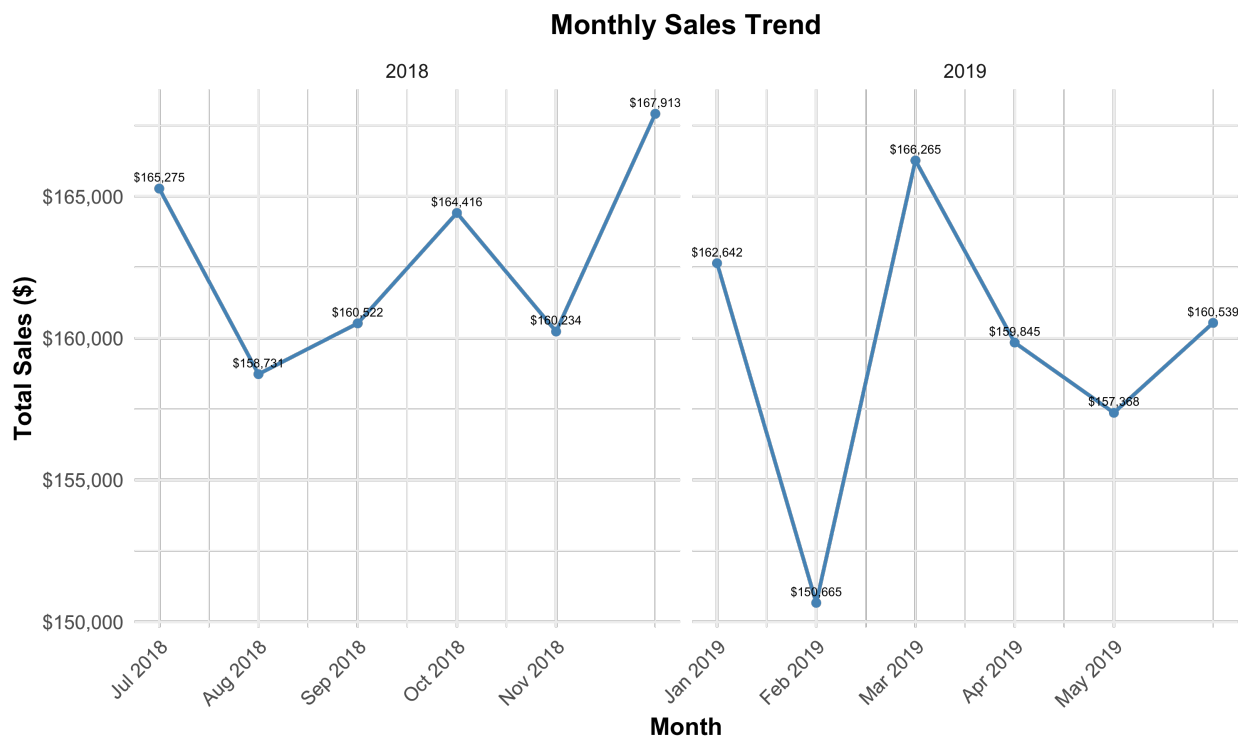
3.1 Top 5 Brands by Total Sales

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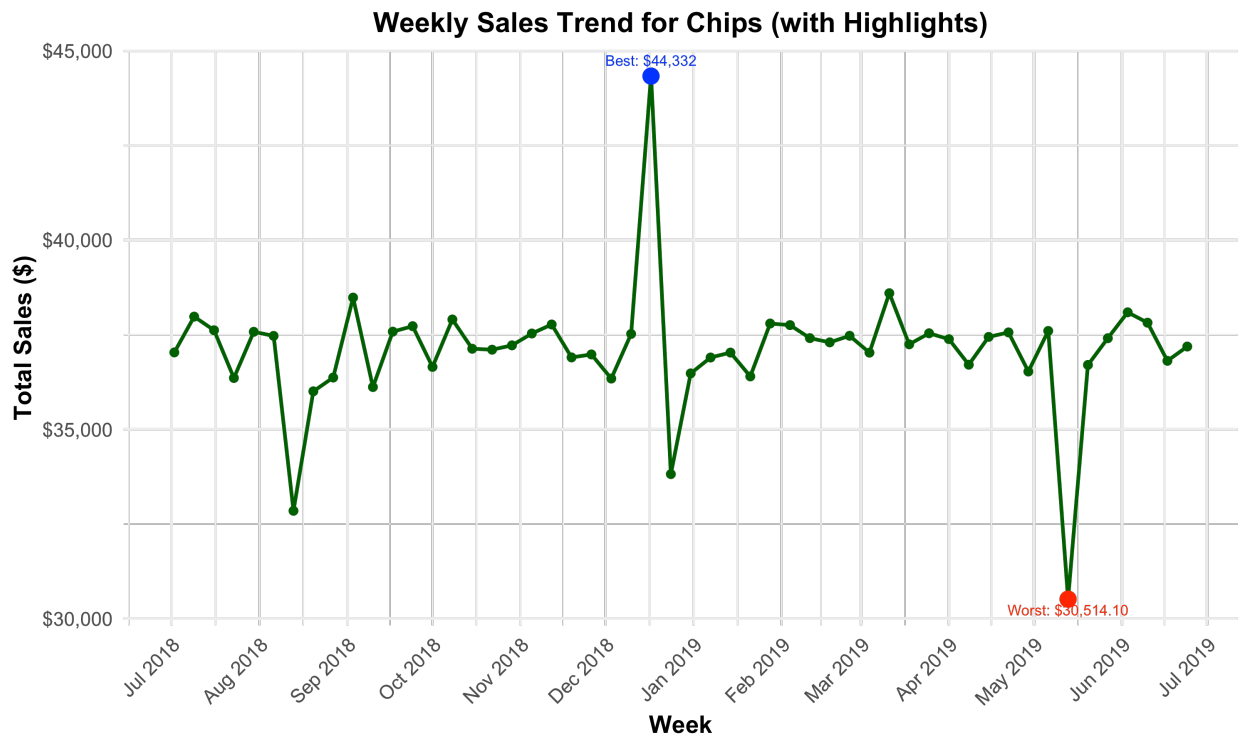
Monthly Sales Trends

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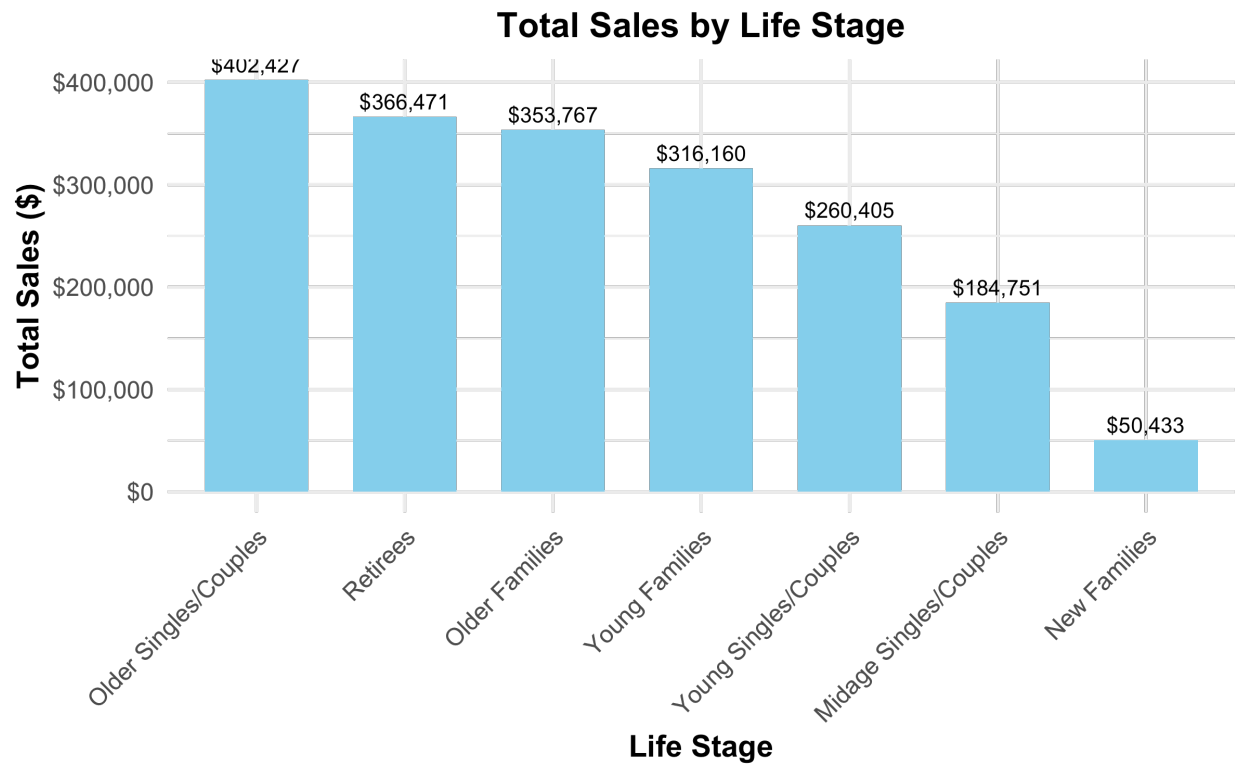
Weekly Sales Trend

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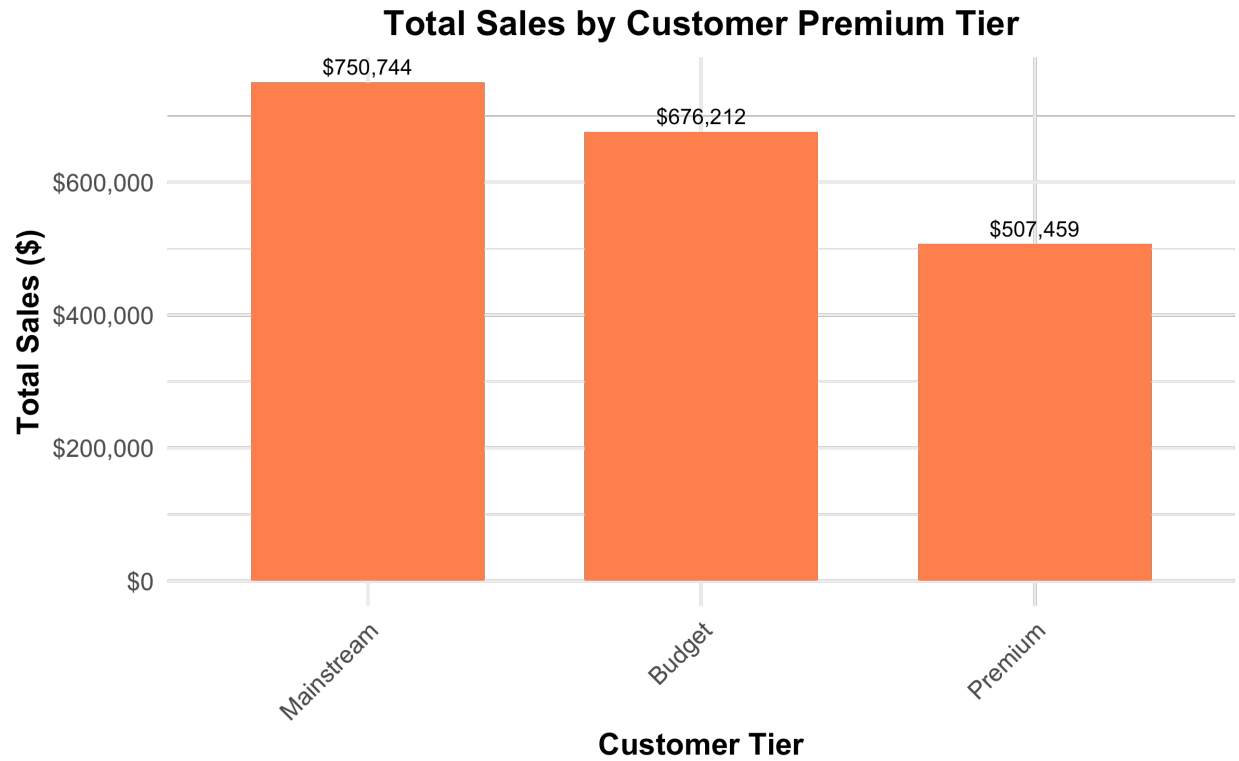
Sales by Life Stage

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3.2 Sales by Customer Premium Tier

```
knitr::include_graphics("outputs/charts/sales_by_customer_premium_r.png")
```



4 Conclusion

Based on the analysis, mainstream customers and holiday months (November–December) present the strongest revenue opportunities. Targeted marketing toward older demographic life stages and premium segment promotions could significantly boost future chip category sales.

5 Appendix

Data source: Quantum transaction and customer data Analysis conducted using R, ggplot2, and RMarkdown
Report prepared by Tianyi Sun